

cooling fan, all assembled) can be powered from a 12V, 6A+ switch-mode power supply (SMPS) unit/module, like the one shown in Fig. 3. Else, try a 12V/7Ah SMF battery. If everything is okay, you will see traces of frost appearing on the plate within seconds.

Note that the primary function of a Peltier chip is cooling, and Peltier chips have different power ratings corresponding to how fast the cold side is able to cool down an object. Another factor generally specified is delta-T (dT), which is the maximum difference between temperature on both sides.

Further, Peltier chips don't function as per specifications, unless there is something to help in taking the heat out from the hot side. That's why a beefy heat-sink is called for. It's the ambient air with its temperature where heat is dissipated.

Well, the assembled and tested cooler engine can now be used to build your own mini refrigerator, can cooler or a tiny air-conditioner. A bit of googling will hopefully provide you fascinating ideas on that.

TEC controllers/drivers

Sometimes you need a dedicated TEC controller/driver. Certainly, there are numerous devices available for advanced applications. On eBay, you can find several devices that would do the job. Fig. 4 shows such a feature-rich device, surprisingly with one feedback channel to accept inputs from an NTC thermistor for temperature stabilisation.

A TEC controller regulates the current supplied to the Peltier chip according to the desired temperature of the object and the actual measured object temperature. To be able to control the temperature of the object, you have to place the sensor on the object. Note that it's important to place the sensor as close as possible to the critical point on the object where you need to maintain the desired temperature.

Since fan cooling of the heat-sink reduces thermal resistance from the heat-sink to the ambient air, most high-end TEC controllers have dedicated fan-control outputs backed by pulse-width modulation (PWM) tech-

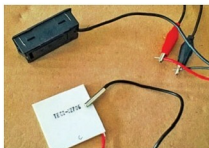


Fig. 2: Testing the TEC chip



Fig. 3: 6A-8A, 12V switch-mode power supply



Fig. 4: Peltier controller sPLC-10

nique. Therefore the fan increases the thermal performance and reduces the temperature difference (dT), allowing use of smaller heat-sinks.

Coefficient of performance

An important measure when selecting a Peltier element is the coefficient of performance (COP). COP is defined as the heat absorbed at the cold side divided by the input power of the Peltier element. The result of a maximum COP is minimum Peltier input power. Thus minimal total heat has to be dissipated by the heat-sink. The lower heat-sink temperature results in a lower dT. So smaller heat-sinks can be used, allowing a more space-saving design. On the other hand,

when optimising costs, a design with a lower COP should be used.

DC or PWM?

There are two power supply/controller modes for thermoelectric coolers to operate by the Peltier effect: DC and PWM. Although in many situations PWM is used to drive Peltier elements, most Peltier element manufacturers suggest DC mode and explicitly advise against direct PWM control of Peltier elements.

It's reported that Peltier elements controlled by PWM are always less efficient than applications controlled by DC. Another issue with PWM mode is electro-magnetic interference (EMI) in the wiring to the Peltier element.

Some experts recommend the use of PWM with an L-C filter to get a clean drive current at higher frequencies, while others prefer the comparatively simple DC mode. Anyway, according to documentations, it is important that the drive current is constant and smooth with very low ripple and noise to achieve good stability. Ripples reduce cooling capability of the Peltier element.

Linear or SMPS?

There are two popular solutions to generate the required DC for driving Peltier elements—linear and SMPS. Since Peltier elements/linear power supply units are driven by direct current, linear power supply units will work optimally but these have a poor efficiency. On the other hand, SMPS units have a high efficiency (> 90%) as their electronic design leads to fewer losses. For this reason, linear power supplies are not recommended to drive Peltier elements.

Author's note

This article shares the essentials and some ideas for the purpose of stimulating imagination and creativity. Readers can purchase most key components from eBay.in, and XK241 2DC SMPS module and SPLC-10 Peltier controller from markets abroad. **Easy**



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